

Happy Engineer's Day

More examples of problems involving **monotonic search space**

Q1

N tasks

K workers

times[] = { 3, 5, 7, 6, 2, 10 }
(to execute all tasks)

$\begin{matrix} T_1 & T_2 & T_3 & & & T_6 \\ \downarrow & \downarrow & \downarrow & \dots & & \downarrow \end{matrix}$

N=6, **K=3**

Conditions -

- Each worker will execute a **Set of continuous tasks**
- All workers start at same time (work in parallel)

output - min **time** required to finish all the tasks

Example $\Rightarrow$

$\begin{matrix} w_1 & w_2 & w_3 \\ 3, 5, 7 & 6 & 2, 10 \end{matrix}$
 $\max(8, 13, 12) = 13$



$\begin{matrix} (3) & (5) & (7) & (6) & (10) \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ w_1 & w_2 & w_3 & w_4 & w_5 \end{matrix}$

$\max(3, 5, 7, 6, 10)$

$\boxed{(3) (5) (7) (6) (10)}$
 w_1

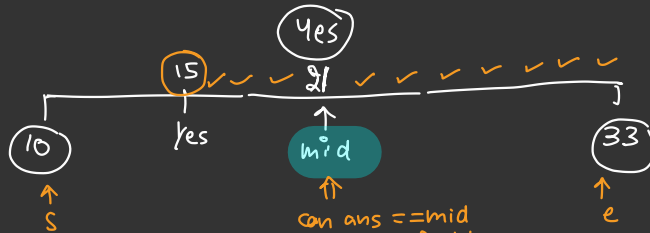
$$\max(15, 6, 12) = 15$$

Min time
acc to
the
best split

Search Space

Min Time = Time to execute largest task

Max Time = Sum of all tasks

$$S = 10$$
$$e = 33$$


Can all tasks be finished in ≤ 21 mins.

$$e = mid - 1$$

$a \rightarrow S=10, e=20, m'd=15$

≤ 15 mins

$$l = \text{mid} - 1$$

3, 5, 7, 6, 2, 10

$$\begin{cases} w_1 = \underline{3} + \underline{5} + \underline{7} + \underline{6} = 21 \\ w_2 = 2 + 10 = 12 \\ w_3 = 0 \end{cases}$$

$$w_1 = 3 + 5 + 7 \leq 15 \text{ mins}$$

b $\rightarrow$ $s=10, e=14, mid=12$

≤ 12 mins
 $\hookrightarrow$ Not possible

$s = mid + 1$

c $\rightarrow$ $s=13, e=14, mid=13$

$\rightarrow$ (yes) $ans = 13$
 $e = mid - 1$

$e=12, s=13$

Stop

$w_2 = 6 + 2 + 10 = 18$ mins ≤ 15
 $w_3 = 10$ mins $= 10$ mins

$\overline{3}, \overline{5}, \overline{7}, \overline{6}, \overline{2}, \overline{10}$

$w_1 = 3 + 5 = 8$

$w_2 = 7$

$w_3 = 6 + 2$

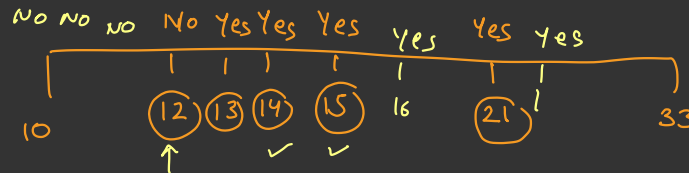
$w_4 = ?$

3, 5, 7, 6, 2, 10

$w_1 = 3 + 5$

$w_2 = 7 + 6$

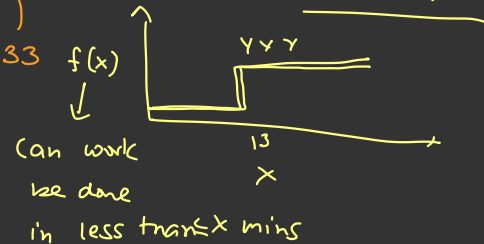
$w_3 = 2 + 10$



No - 0

Yes - 1

non-dec fn





$$K=3 \left[\begin{array}{l} w1 = 3 + 5 + 7 + \cancel{6} \\ w2 = 6 + 2 + \cancel{10} \\ w3 = 10 \end{array} \right]$$

limit
15

← 15 Mins, 20, 28, 40
Yes

Code

minTime (tasks[], N, K) {

s = max (tasks)

e = sum (tasks)

ans = e

while (s <= e) {

mid = (s + e) / 2

$O(N) \rightarrow$ if (canExecute (tasks, N, K, mid)) {

ans = mid

e = mid - 1

}

else {

s = mid + 1

~

}

print (ans);

}

TC

$O(n \log (\text{sum of tasks}))$

Space

$O(1)$

Can execute (tasks[], N, K, timeLimit) {

$\Rightarrow w = 0$
cnt = 1

for (i = 0; i < N; i++) {

if ($w + \text{tasks}[i] \leq \text{timeLimit}$) {

$\rightarrow w = w + \text{tasks}[i]$

3
else {

$\rightarrow w = \text{tasks}[i];$

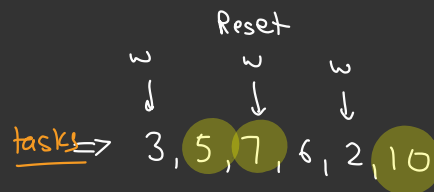
cnt++

if (cnt > K) { return false },

3
}

return true.

}



N = 6

K = 3

timeLimit = 13

① $w = 0 + 3 + 5$

$8 + 7 \leq 13$

② $w = 7 + 6$

$7 + 6 \leq 13$

③ $w = 2 + 10$

$13 + 2 \leq 13$

"Google Interview Problem"

(45 min) Standard Time.

→ Multiplayer game with K players including you

Coins =

10	3	6	5	8
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→ Your goal is to cut the array into K segments

Each player gets a segment and you get the smallest segment

10015
SS
→ Think time

Value of segment = Sum of coins in that segment

Maximise the value of smallest segment you will get
(by value)

Sample Input

10 | 20 | 15 | 5 | 8
= (10) (35) (13)

$K=3$

You $\Rightarrow$ 10 $\leftarrow$

10 | 20 | 15 | 5 + 8
(30) (15) (13)

You $\Rightarrow$ (13) $\leftarrow$

$S = \min(\text{coins})$

$e = \text{sum}(\text{coins})$

while ($S \leq e$)

$\text{mid} = (S + e) / 2$

if (canGive(mid , coins, K)) {

$\text{ans} = \text{mid};$

$S = \text{mid} + 1$

}
else {

$e = \text{mid} - 1$

}

return ans.

}

10, 20, 15, 5, 8

$K=3$

no no
↑ ↑
 $S=5$ 17 $\text{mid}=(31)$ $e=58$

I can get 31 coins if everyone
gets ≥ 31 coins

$F1 = 10 + 20 + 15$
 $F2 = 13$
 $F3 =$

$10 + 20 = 30 \geq 18$

$15 + 5 = 20 \geq 18$

8 8 ≥ 18

$S=5$ $e=17$ $\text{mid}=(11)$ $10 + 20 = 30 \geq 11$

15 ≥ 11

13 ≥ 11

yes
↑
 $S=5$ $\text{mid}=11$ $e=17$

$S=12$ $e=17$

$\text{mid}=(14)$

bool can Give (mid, coins, k) {

cnt = 0

S = 0

```
for (i = 0; i < n; i++) {  
    if (S + coins[i] ≥ mid) {  
        → cnt++;  
        → S = 0 // Reset  
    }  
    else {  
        S = S + coins[i],  
    }  
}
```

⇒ if (cnt ≥ k) { return True }
 else { return false }

}

(10, 20, 15, 5, 8)

mid = 13

at least

mid

coins

to everyone

S1 = 0 + 10 + 20 ≥ 13

(20)

S2 = 15 ≥ 13

(15)

S3 = 5 + 8 ≥ 13

(13)

(0)

S = 0 + 10 ≥ 13
= 10 + 20 ≥ 13

(S2)

S = 0 + 15 ≥ 13

(S3)

S = 0 + 5 ≥ 13
= 5 + 8 ≥ 13

(S3)

S = 0

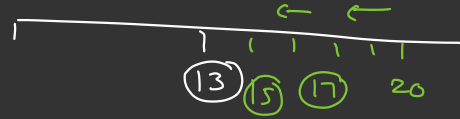
$10, 20, 10, 5, 8, 9$

$\uparrow$
30

$\uparrow$
15

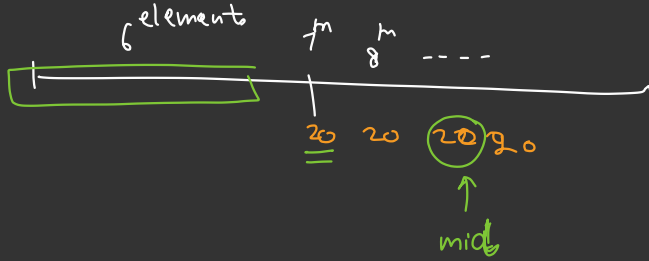
$\uparrow$
17

≥ 13



$\Rightarrow$ Painter's Partition
 $\Rightarrow$ Reading Books

$\left. \begin{array}{l} \Rightarrow \text{Painter's Partition} \\ \Rightarrow \text{Reading Books} \end{array} \right\} \rightarrow \text{same as prob \#1}$
(Google Search)



7^m elem
 $\downarrow$
 $20 (-----)$

8^m
 element

if (cnt <= $N/2$) {

ans = mid ;

s = mid + 1 ;

}

